

1 Oct 20-22

1,2,4,6,7

$$1: \Delta\nu = \frac{c}{2\mu L} = \frac{3*10^8}{1} = 3*10^8 \text{ Bec} : \Delta\nu = 6*10^8$$

$$\text{So : } n = \frac{\Delta\nu}{3*10^8} = 2.$$

q could be : q-1;q;q+1.

$$\text{and } \nu = \frac{qc}{2\mu L} \text{ so } q = \frac{2\mu L\nu}{c} = \frac{5.85*10^{14}}{3*10^8} = 1.95*10^{16} \text{ and } q+1 \text{ q-1.}$$

2:

$$\Delta\nu = \frac{3*10^8}{2} = 1.5*10^8 = 0.1* \Delta\nu$$

so n=10 ,q could have 11 . if q only neede 1 ,so

$$\Delta\nu = 1.5*10^9 = \frac{3*10^8}{2L} \text{ so } L=0.1 \text{ m .}$$

4:

$$(1): P/S = \frac{50W}{(25\mu m)^2 \pi} \mu m^2 = K$$

$$(2): \frac{K}{10^4} = N_1; \frac{K}{10^3} = N_2$$

6:

$$\theta = \frac{1.22*632.8nm}{2mm} = \theta_1$$

$$\theta_0 = \frac{\lambda}{\pi\omega_0} = \frac{\lambda}{\pi\sqrt{\frac{\lambda L}{2\pi}}}$$

7:

$$\omega_0 = \sqrt{\frac{\lambda L}{2\pi}}$$

$$\omega(z) = \omega_0 \sqrt{1 + \frac{\lambda z^2}{\pi\omega_0^2}}$$

$$L = 0.4; \lambda = 0.6328\mu m; z = 56cm$$

9:

$$R = z[1 + (\frac{f}{z})^2]^{\frac{3\sqrt{3}}{7}} = f, R'_1 = R'_2 = \frac{6\sqrt{3}}{7}$$

$$\omega_0 = \sqrt{\frac{\lambda f}{\pi}} = \sqrt{\frac{\lambda 3\sqrt{3}}{7\pi}}$$

$$\omega = \omega_0 \sqrt{1 + (\frac{f}{z})^2} = \sqrt{\frac{\lambda 3\sqrt{3}}{7\pi}} \sqrt{1 + (\frac{3\sqrt{3}}{7z})^2}$$

$$\frac{R(z_1)-z_1}{z_1} = (\frac{f}{z_1})^2; \frac{R(z_2)-z_2}{z_2} = (\frac{f}{z_2})^2$$

$$\frac{1.5-z_1}{3+z_1} = \frac{1-z_1}{z_1}; z_1 + z_2 = 1; \frac{6}{7} = z_1; z_2 = \frac{1}{7};$$

10:

$$2 = 1[1 + f^2]; f=1; \omega = \omega_0 \sqrt{1 + (\frac{f}{z})^2} = \sqrt{\frac{2\lambda}{\pi}} = 2.5977mm$$

$$\omega = \omega_0 \sqrt{1 + (\frac{f}{z})^2} = \omega_0 \sqrt{1 + f^2} = \omega_0 \sqrt{R} = \sqrt{\frac{\lambda \sqrt{R-1} R}{\pi}} = 0.3$$

$$0.3^2 \pi / \lambda = R \sqrt{R-1}; R=41.7733cm.$$

$$\omega_0 = \sqrt{\frac{\lambda \sqrt{R}}{\pi}}$$

$$\omega = \omega_0 \sqrt{1 + f^2}; R = 1 + f^2; f = \sqrt{R-1}; \omega_0 \sqrt{R} = \sqrt{\frac{\lambda R \sqrt{R}}{\pi}}$$

$$R = (0.3^2 \pi / \lambda)^{2/3} =$$

$$f = \frac{\sqrt{2(R-2)^2(2R-2)}}{2R-4} = \frac{2\sqrt{R-1}}{2} = \sqrt{R-1}$$

$$\omega_0 = \sqrt{\frac{\lambda \sqrt{Rz}}{\pi} \frac{R}{z}}; f = \sqrt{(\omega_0^2 \pi / \lambda)^{2/3}}$$

$$(\omega_0^2 \pi / \lambda)^{2/3} = R$$

$$\omega = \omega_0 \sqrt{1 + (\omega_0^2 \pi / \lambda)^{2/3}}$$

2 sketches